

L6 – Snap Fit Speaker Case

Objective

This week I had to parametrically design a small part that snap fits onto a feature of an artifact we measured in class. The point was to use tolerances, allowances, and the fit classes from lecture (RC, LC, LT, LN, and FN) to size the mating features, then print the part and test the fit. My artifact is a small oval speaker, and my part is a case that holds it in place with four snap fit lips.

Design Idea

The case is a hollowed out rectangular box. The speaker drops into the open end, and four small lips near the opening catch the speaker's front flange so it can't fall back out. I'm not using the speaker's four mounting holes, so nothing in the design touches them.

- **Internal cavity:** matches the speaker's width and length, plus an allowance on each side.
- **Walls:** 1.6 mm thick.
- **Lips (x4):** two on each long wall. Each lip is 5 mm wide, 1 mm tall, and sticks out 1 mm from the wall. The side the speaker enters from has a 45° chamfer so the flange can push past it. The underside is flat, so once the speaker is snapped in, the flat face sits on top of the flange and holds it. The lips stay away from the electronics on the back of the speaker.
- **Cable hole:** a half-round cutout with a 5 mm radius, cut through the back of the case at one short end, like a mouse hole. I cut off the JST style plug, so the wires are bare and pass through the hole easily.

Research: Tolerances, Allowances, and Fits

Definitions

Machinery's Handbook (31st Ed., p. 640) defines **tolerance** as the amount of variation permitted on the dimensions or surfaces of machine parts. It defines **allowance** as a prescribed difference in dimensions used to get different classes of fits between parts. So tolerance is how much a single dimension is allowed to wander, and allowance is the gap (or overlap) I choose on purpose between two mating parts.

Which Fit Class Is This?

The speaker sits in the case like a **locational clearance fit (LC)**. The parts have to go together freely by hand, but they should still locate each other without a lot of play. The lips are a different story. During insertion the flange has to push past a 1 mm lip, which is a temporary interference, and after that the lip springs back and holds the speaker. So the overall behavior is LC for the pocket, with the snap fit lips supplying the retention. It is not an LN or FN fit, because nothing is pressed together permanently.

What the Handbook Says About My Sizes

Table 6 (ANSI/ASME B4.1-1967 (R2020), Machinery's Handbook 31st Ed., p. 650) gives standard tolerances in thousandths of an inch, by nominal size range and tolerance grade. I converted my measurements to inches (1 in = 25.4 mm) and pulled the row for each one.

Feature	Size	Table 6 row (in)	Grade 7	Grade 12	Grade 13
Width	28.28 mm = 1.113 in	0.71 to 1.19	0.0008 in = 0.020 mm	0.008 in = 0.20 mm	0.012 in = 0.30 mm
Length	39.86 mm = 1.569 in	1.19 to 1.97	0.0010 in = 0.025 mm	0.010 in = 0.25 mm	0.016 in = 0.41 mm
Depth	13.34 mm = 0.525 in	0.40 to 0.71	0.0007 in = 0.018 mm	0.007 in = 0.18 mm	0.010 in = 0.25 mm

The precise grades a locational fit is built on are down around 0.02 to 0.06 mm for these sizes. My printer can't hold anything close to that. Table 7 in the handbook relates machining processes to tolerance grades, and 3D printing isn't on it. The closest processes it lists are milling and drilling, which run roughly from grade 10 to 13. The Bambu Lab A1 holds about 0.2 mm on small features (my working assumption, which the test print ended up confirming), and that lines up with grade 12 in the table: 0.20 mm on the width and 0.25 mm on the length.

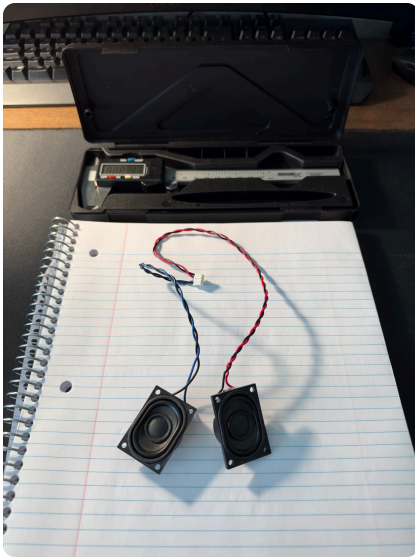
Allowance Decision

I set the clearance to **0.2 mm per side**. That puts the internal cavity at the speaker size plus 0.4 mm in each direction, so the speaker slides in by hand, and it stays inside the grade 12 tolerance band I expect from the printer. It is still a lot bigger than what the handbook expects for an LC fit, but the handbook numbers assume machining, so I used them as the design intent and used the printer's real accuracy to pick the actual number.

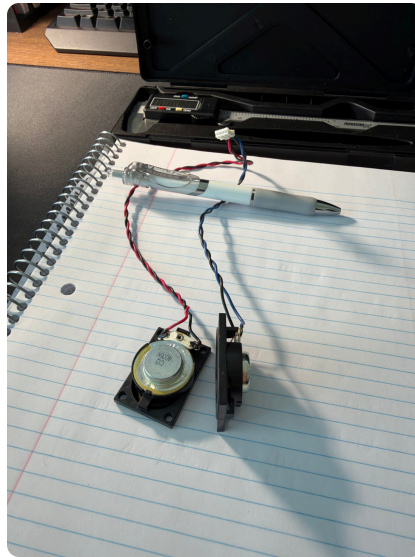
Measuring the Artifact

I measured the speaker with a digital caliper that reads to 0.01 mm. The step depths in the side profile don't matter for this design, since the case is sized to the speaker's overall envelope. The magnet is centered on the rectangle.

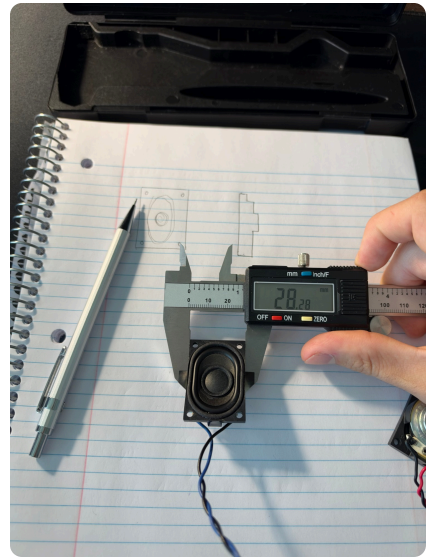
Feature	Measurement
Width	28.28 mm (rounded to 28.3 mm on my sketch and in the CAD model)
Length	39.86 mm
Total depth	13.34 mm
Front flange thickness	3.59 mm
Magnet diameter	27.85 mm



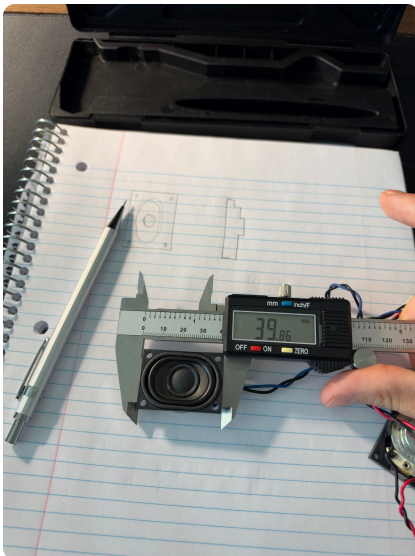
The two speakers, with the digital caliper in the background.



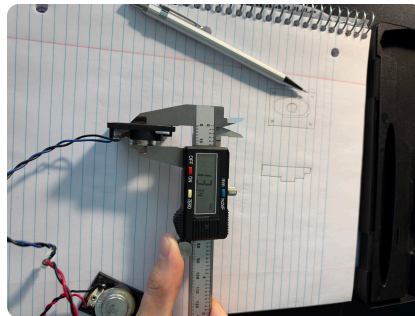
The back of the speaker. The round magnet sits in the center, and the wire terminals are soldered to the back of the frame.



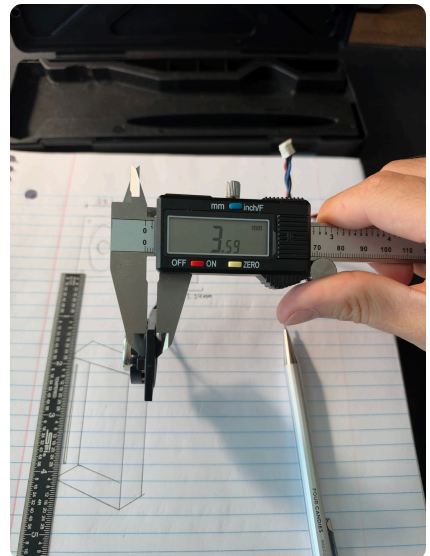
Width: 28.28 mm.



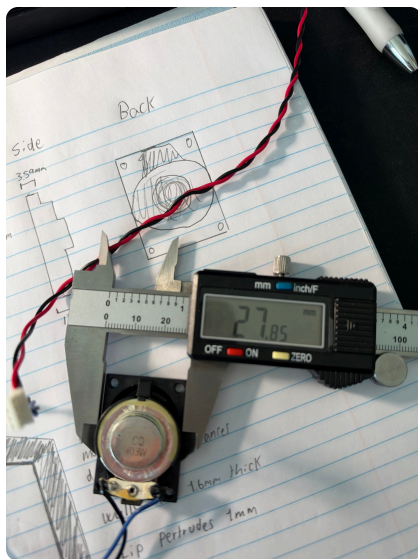
Length: 39.86 mm.



Total depth: 13.34 mm.

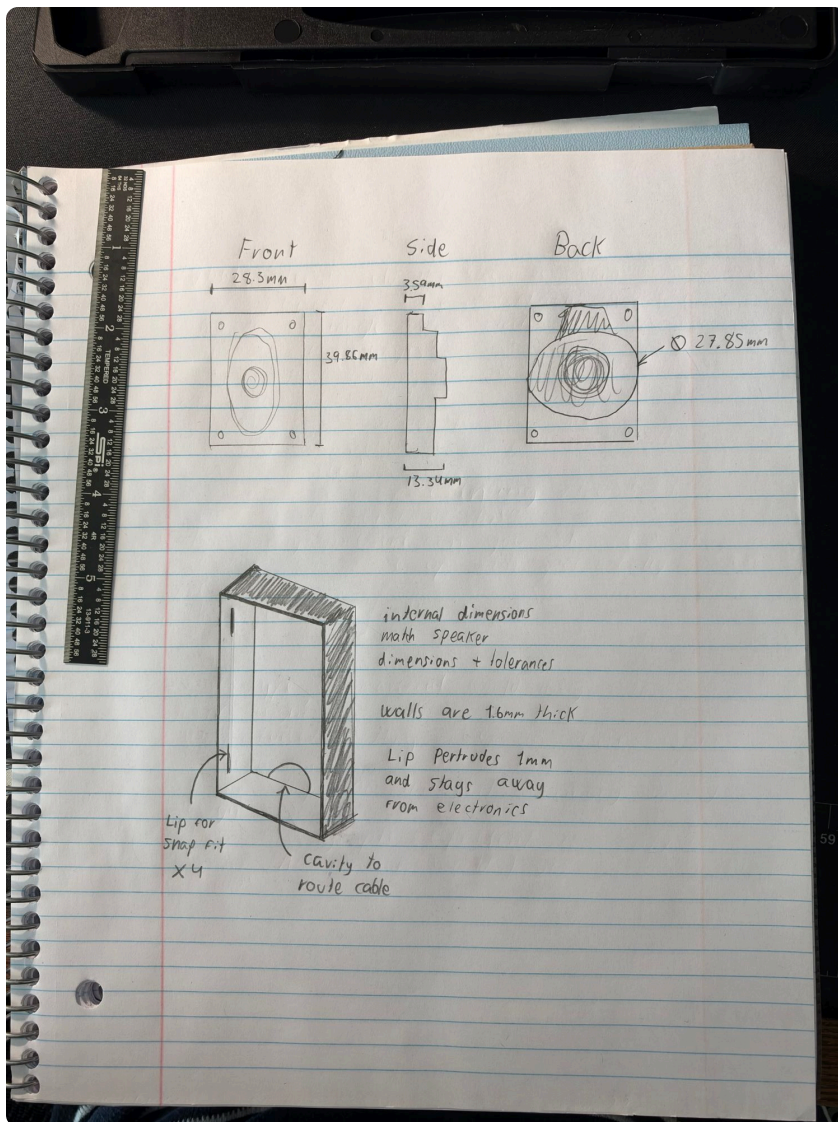


Front flange thickness: 3.59 mm.



Magnet diameter: 27.85 mm.

Hand Sketch



The top row is the speaker's front, side, and back views with the measurements written on. The bottom sketch is the case concept: internal dimensions that match the speaker plus tolerances, 1.6 mm walls, four lips that protrude 1 mm and stay away from the electronics, and a cavity to route the cable.

Parametric Design

Parameters Used

I built the case around a small set of parameters so that changing the speaker size or the clearance updates the whole model.

Parameter	Value	Why
Speaker width	28.3 mm	Measured width of the speaker (28.28 mm, rounded)
Speaker length	39.86 mm	Measured length of the speaker
Speaker depth	13.34 mm	Measured total depth of the speaker
Clearance	0.2 mm per side	Printer allowance, chosen from Table 6 and the A1's accuracy
Wall thickness	1.6 mm	Stiff enough to hold the speaker, thin enough to flex at the lips
Lip width	5 mm	Small enough to stay clear of the electronics
Lip height	1 mm	Thin catch that sits on the speaker's flange
Lip protrusion	1 mm	How far each lip reaches over the front flange
Lip offset	5 mm	Distance from the end of the wall to the start of each lip
Cable hole radius	5 mm	Big enough for the bare wires to pass through

The rest of the model follows from those. The outer size of the case is the speaker size plus 0.4 mm of clearance plus 3.2 mm of wall (two walls), which comes out to **31.90 mm × 43.46 mm**. The internal cavity is 28.7 mm × 40.26 mm. The case is 16.94 mm tall, which works out to the 13.34 mm speaker depth, plus 0.4 mm of clearance, plus 3.2 mm of wall.

TO ADD: screenshot of the SolidWorks equations or global variables list, and the variable names you used.

Why These Parameters

The three speaker dimensions are the ones I measured, so if I swap in the other speaker or remeasure, the case follows. The clearance is a parameter on its own because it was the number I was least sure about, and I wanted to be able to change it in one place after the test print. Wall thickness and the lip sizes are parameters because they control how stiff the case is and how hard the speaker is to push in.

Did the Values Change?

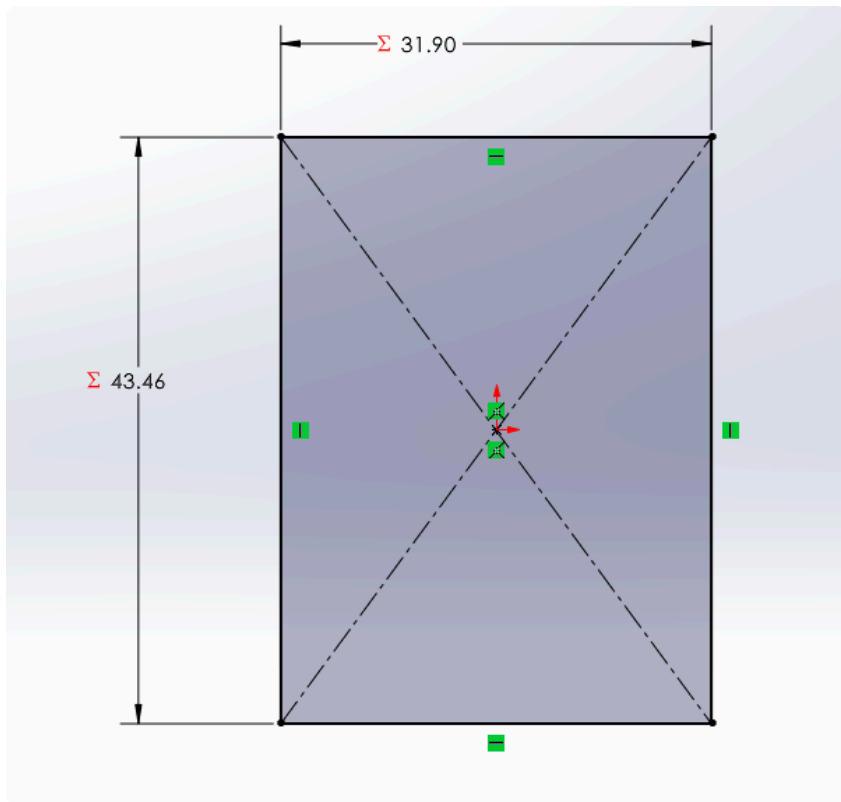
- **Clearance:** I started with 0.2 mm per side, briefly looked at 0.4 mm per side to be safe with the printer, then went back to 0.2 mm. At 0.4 mm the allowance is well past what the handbook table expects for a locational fit, and it would have been harder to justify. 0.2 mm keeps the design closer to the table and it matches grade 12 for the printer.
- **Lip relief cuts:** I first planned cuts above and below each lip so the wall could flex more easily. I left them out of the final model, because the walls are thin enough that they already give me the flex I need.

Lip Design

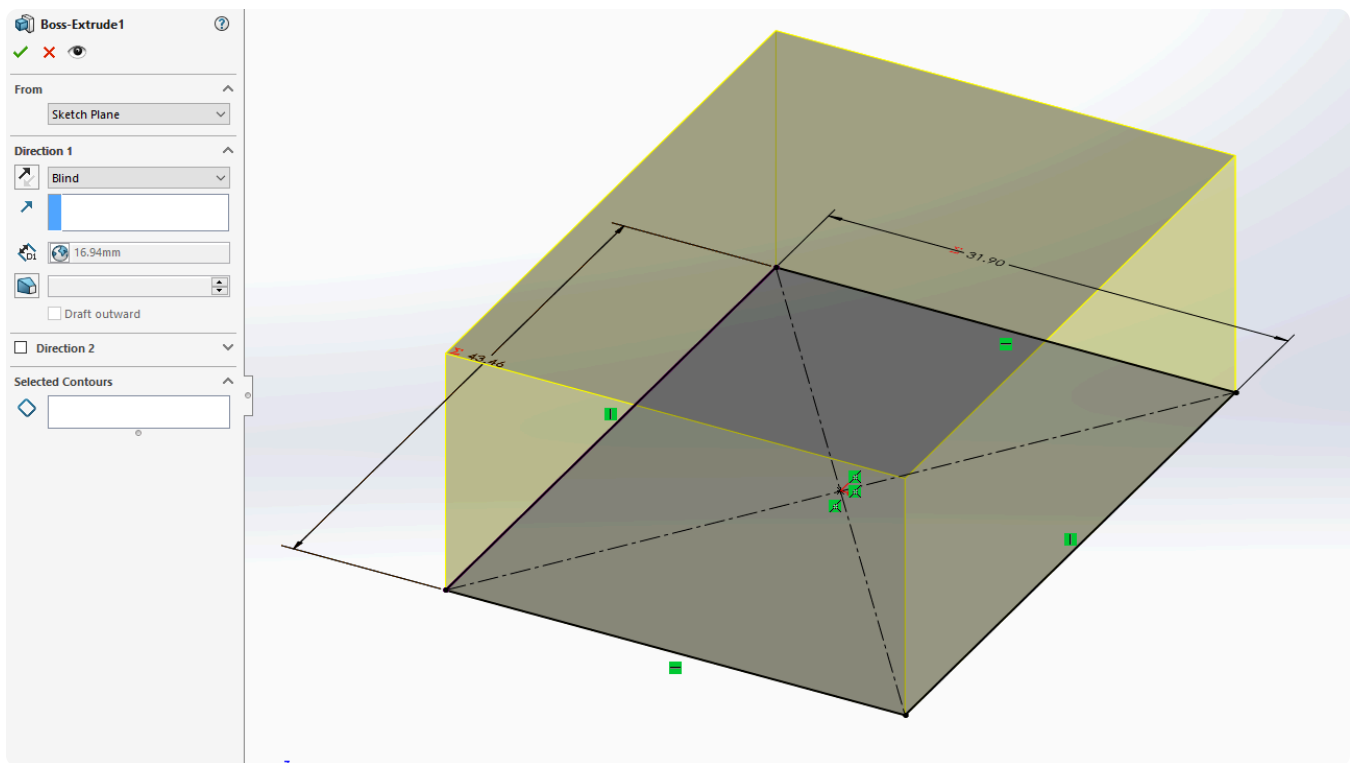
Each lip is 5 mm wide, 1 mm tall, and sticks out 1 mm from the wall. The entry side has a chamfer of 1 mm at 45°, which is the full protrusion of the lip, so the entry face is a smooth ramp for the flange to ride up. The other side is flat, so the lip can't slide off once it's on top of the flange. I didn't do any beam calculations for this assignment. The lip's job is only to hold a light speaker in place.

CAD Process

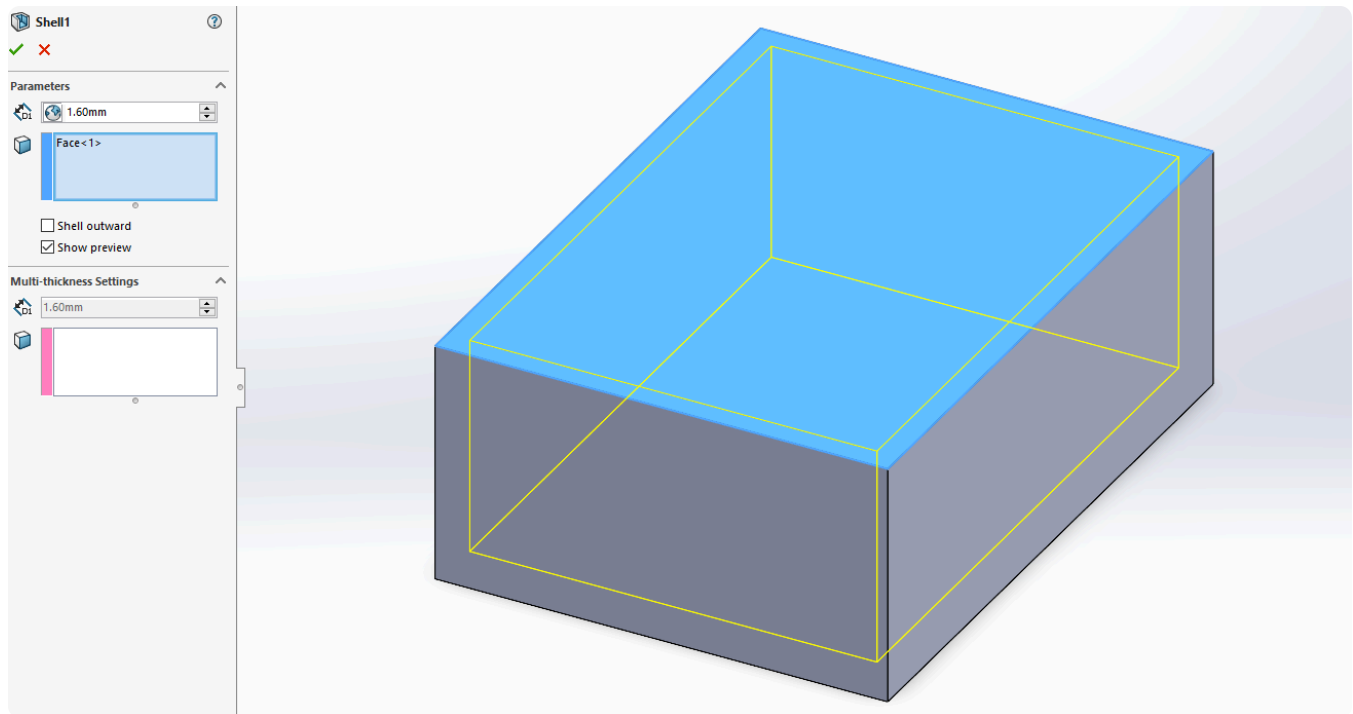
I modeled the case in SolidWorks. The dimensions with a red sigma symbol in the screenshots are driven by equations, so they follow the parameters above. The green symbols in the sketches are constraints.



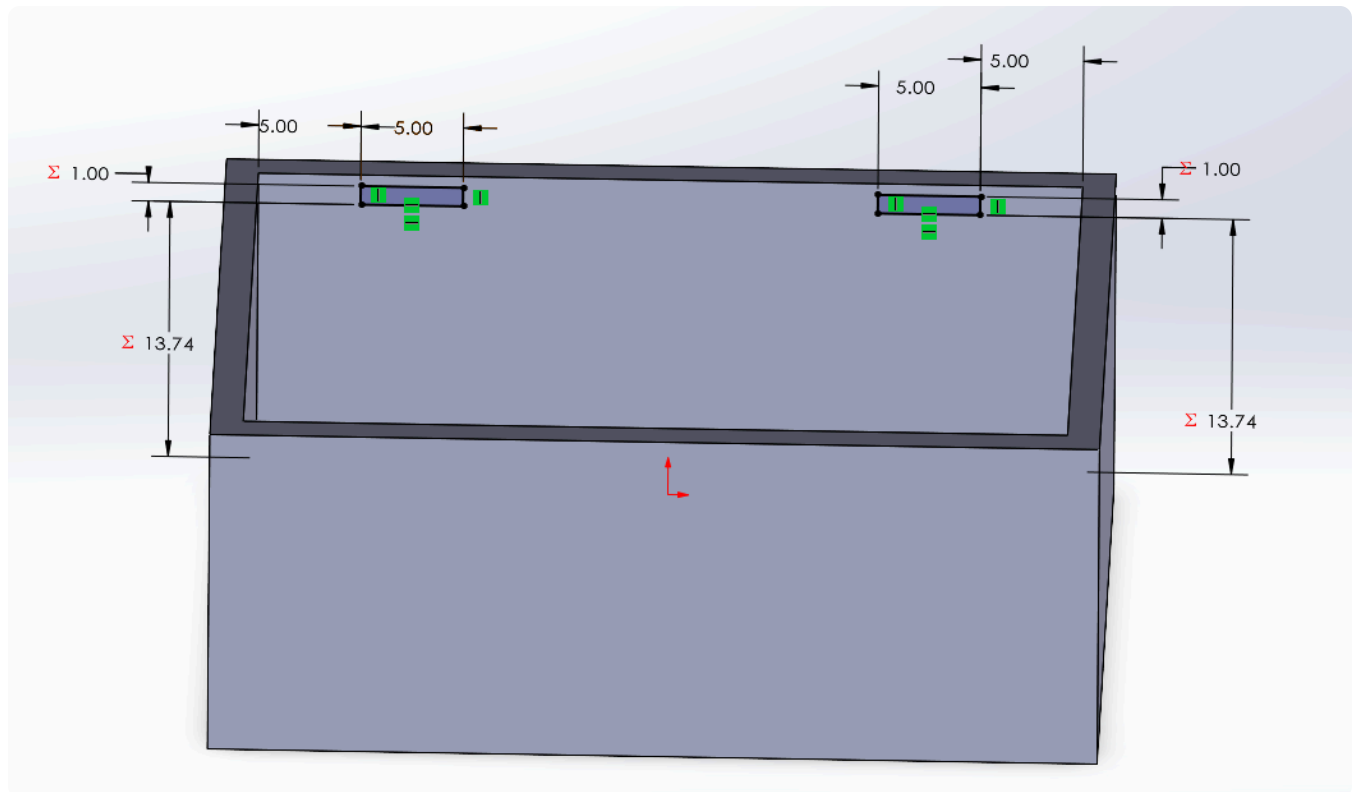
Step 1, sketch the box: I sketched a rectangle centered on the origin, using diagonal construction lines through the corners to center it. It is constrained horizontal and vertical, and the width (31.90 mm) and length (43.46 mm) come from the equations for speaker size, clearance, and wall.



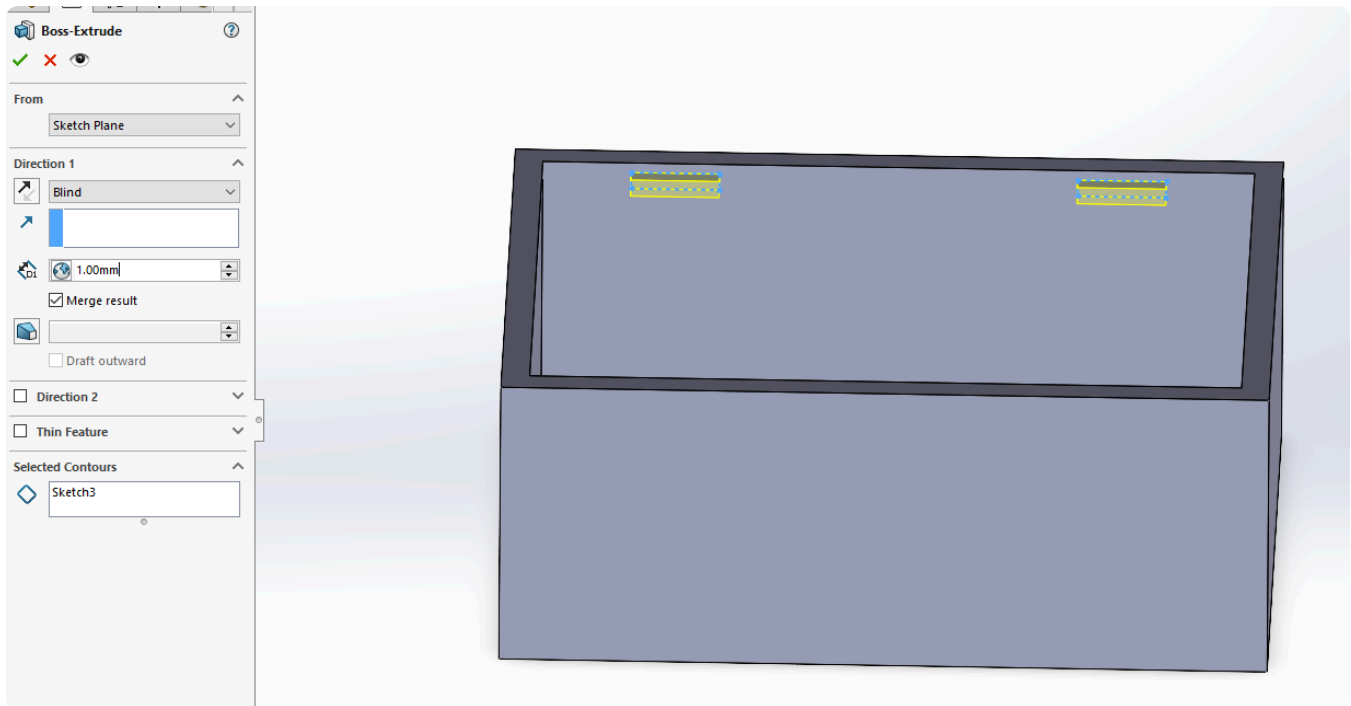
Step 2, extrude the box: Boss-Extrude, blind, 16.94 mm, which is the height of the case.



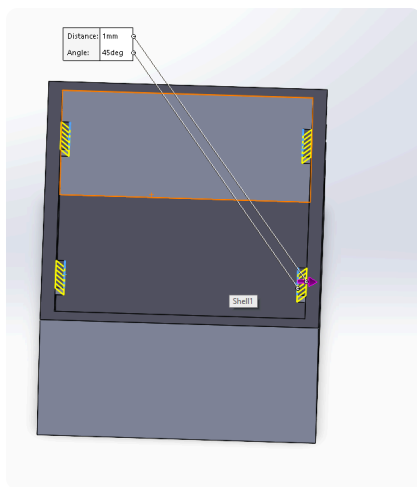
Step 3, shell the box: Shell with a 1.60 mm thickness, removing the top face to make the open end the speaker goes into.



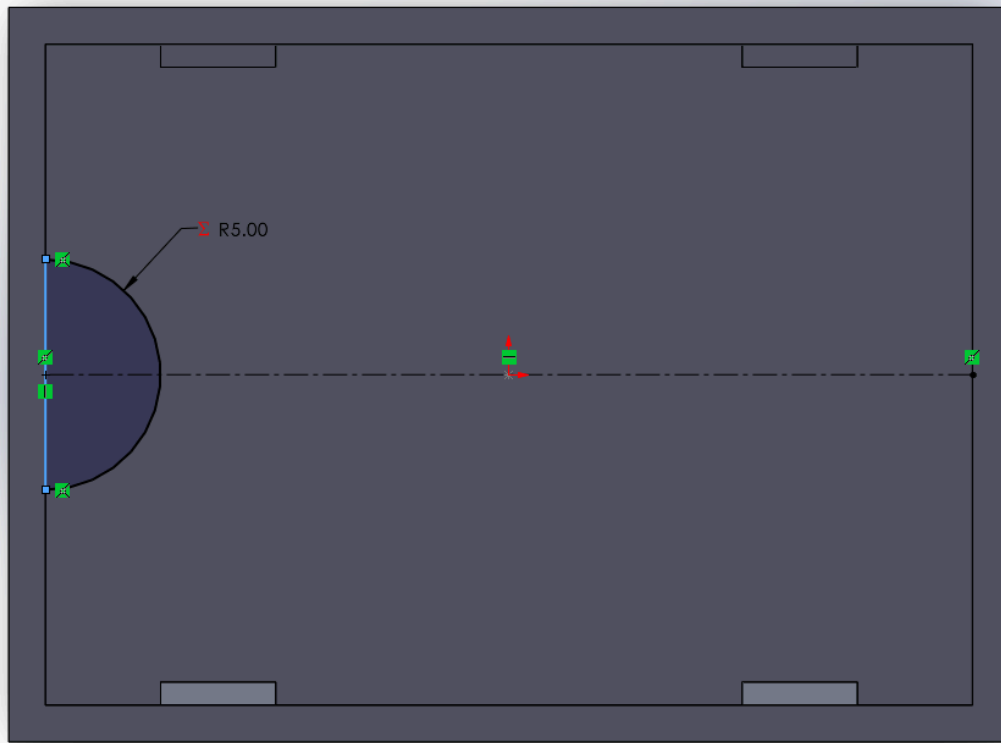
Step 4, sketch the lips: On the inside face of a long wall I sketched two rectangles, each 5.00 mm wide and 1.00 mm tall, 5.00 mm from each end of the wall. Their position is 13.74 mm, which is the speaker depth plus 0.4 mm. The two lips are constrained equal to each other so they always match. **TO ADD: confirm where the 13.74 mm is measured from (the inside floor or the open edge).**



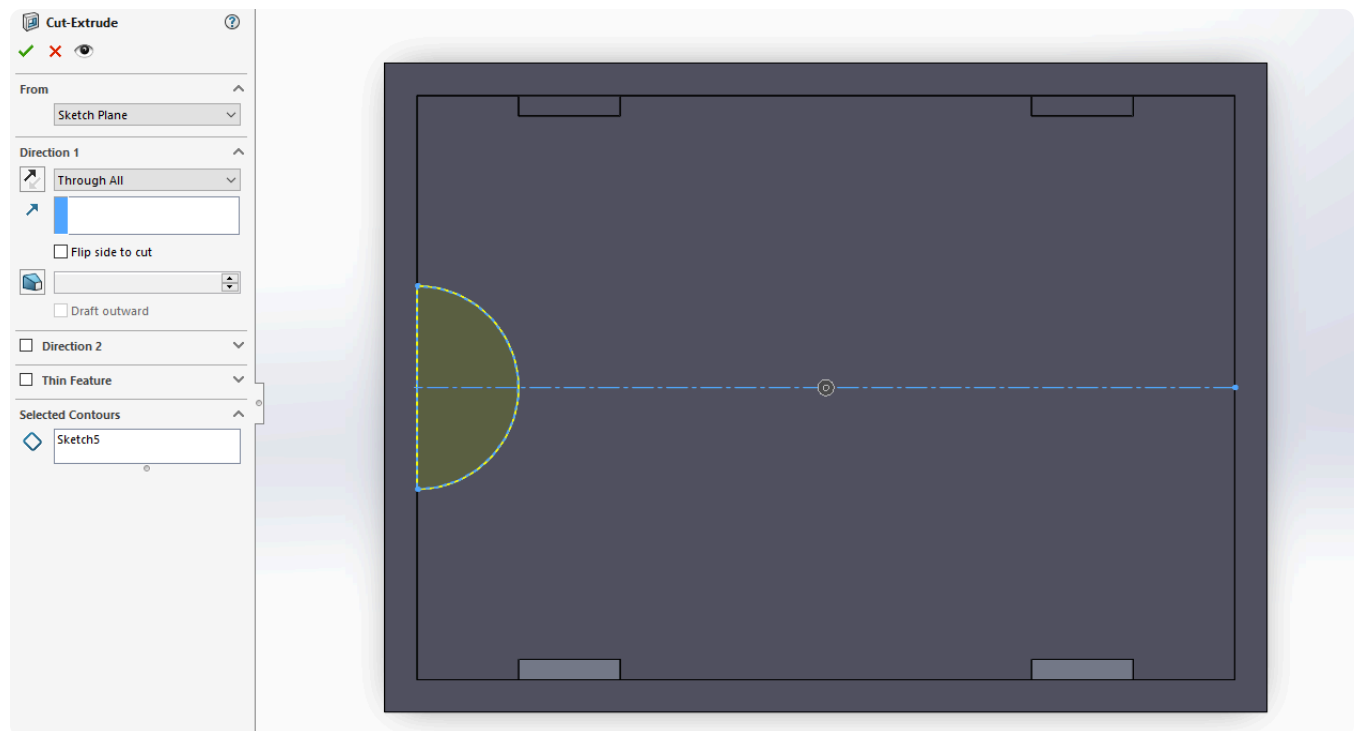
Step 5, extrude the lips: Boss-Extrude, blind, 1.00 mm, with Merge result checked so the lips become part of the case.



Step 6, chamfer the lips: Chamfer with a distance of 1 mm and an angle of 45° on the entry edge of each lip, so the flange can slide over it easily.



Step 7, sketch the cable hole: On the floor of the case I sketched a half circle with a 5.00 mm radius. Its flat side lies on the inside face of the short wall, and it is centered on the centerline of the box. The four lips can be seen along the long walls.



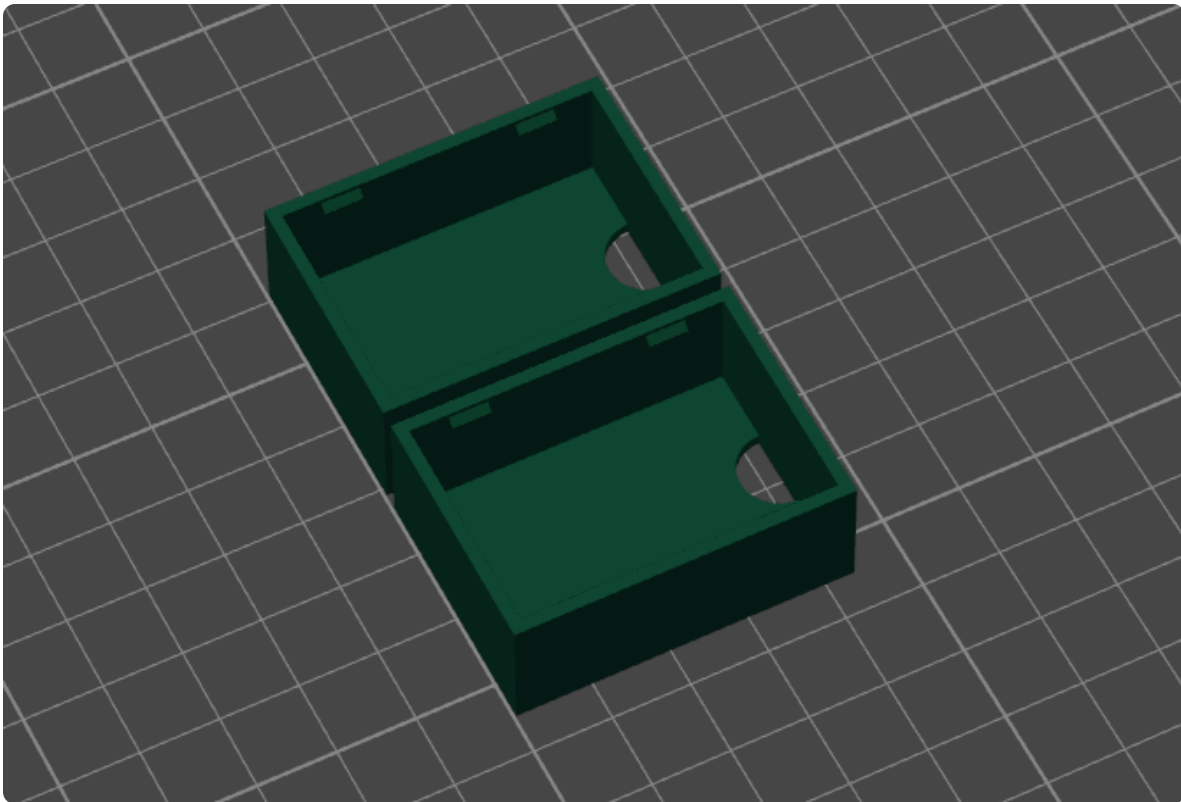
Step 8, cut the cable hole: Cut-Extrude, Through All, which cuts the half circle through the back of the case.

3D Printing

Printer and Layout

I printed on a **Bambu Lab A1** with a 0.4 mm nozzle and a 256 mm build volume, in **Overture Matte PLA (green)**. I printed two cases at once, side by side, one for each of the two speakers. I left them in the layout Bambu Studio recommended.

Each case is 31.90 mm × 43.46 mm × 16.94 mm, which is a bounding box of about 23.5 cm³.



The two cases on the plate in Bambu Studio.

Build Orientation and Supports

The back of the case sits flat on the plate and the hollow part faces up. I chose that orientation because it needs the least amount of support and it prints faster. The only overhangs are the lips, and each one is a 1 mm ledge that is 5 mm wide, so it is small. The chamfer is on the top of the lip, which means it doesn't create an overhang.

In the slicer I turned on tree supports (auto) with a 30° threshold and small overhangs removed. **TO ADD: whether any supports actually printed, why the support size ended up as it**

did, and what you used to remove them.

Layers and Walls

- **Wall thickness:** 1.6 mm in the design. The slicer prints 2 wall loops.
- **Number of layers:** 85 (from the printer's screen).
- **Layer thickness:** 0.2 mm, so 85 layers is 17 mm, which lines up with the 16.94 mm height of the case.
- **Build volume of the print:** 31.90 mm × 43.46 mm × 16.94 mm per case, with two cases on the plate.

Slicer Settings

I sliced in Bambu Studio with the 0.20mm Standard @BBL A1 process.

Printer

Bambu Lab A1

Texture...

Sync info

Nozzle

Diameter

Flow

Project Filaments

+
 -

1

Process

Global
 Objects

Advanced
 ☒

Quality
 Strength
 Speed
 Support
 Others

Layer height

Layer height	0.2	mm
Initial layer height	0.2	mm

Line width

Default	0.42	mm
Initial layer	0.5	mm
Outer wall	0.42	mm
Inner wall	0.45	mm
Top surface	0.42	mm
Sparse infill	0.45	mm
Internal solid infill	0.42	mm
Support	0.42	mm

Printer, filament, and quality: Bambu Lab A1, 0.4 mm nozzle, Overture Matte PLA, process 0.20mm Standard @BBL A1. Layer height and initial layer height are both 0.2 mm. Line widths are 0.42 mm for the default, outer wall, top surface, and internal solid infill, 0.45 mm for the inner wall and sparse infill, and 0.5 mm for the initial layer.

Walls

Wall loops	2
Detect thin wall	☐

Top/bottom shells

Top surface pattern	 Monotoni...
Top shell layers	5
Top shell thickness	1 mm
Top paint penetration layers	5
Bottom surface pattern	 Monotonic
Bottom shell layers	3
Bottom shell thickness	0 mm
Bottom paint penetration layers	3
Internal solid infill pattern	 Rectilinear

Sparse infill

Sparse infill density	15 %
Sparse infill pattern	 Grid
Length of sparse infill anchor	400% mm or %
Maximum length of sparse infill anchor	20 mm or %

Walls and shells: 2 wall loops, 5 top shell layers (1 mm), and 3 bottom shell layers (0.6 mm). The bottom and top shells add up to 1.6 mm, which is the thickness of the back of the case, so the back plate prints solid.

Infill: 15 percent, grid pattern. The case is mostly thin wall, so the infill barely shows up.

Initial layer speed

Initial layer	50	mm/s
Initial layer infill	105	mm/s

Other layers speed

Outer wall	200	mm/s
Inner wall	300	mm/s
Small perimeters	50%	mm/s or %
Small perimeter threshold	0	mm
Sparse infill	270	mm/s
Internal solid infill	250	mm/s
Vertical shell speed	80%	mm/s or %
Top surface	200	mm/s

Speed: initial layer 50 mm/s and initial layer infill 105 mm/s. Outer wall 200 mm/s, inner wall 300 mm/s, sparse infill 270 mm/s, internal solid infill 250 mm/s, and top surface 200 mm/s. A slow first layer helps the case stick to the plate.

Support

Enable support



Type	tree(auto)
Style	Default
Threshold angle	30°
On build plate only	☐
Support critical regions only	☐
Remove small overhangs	

Supports: enabled, tree (auto), threshold angle 30°, remove small overhangs on, 0 raft layers.

Raft

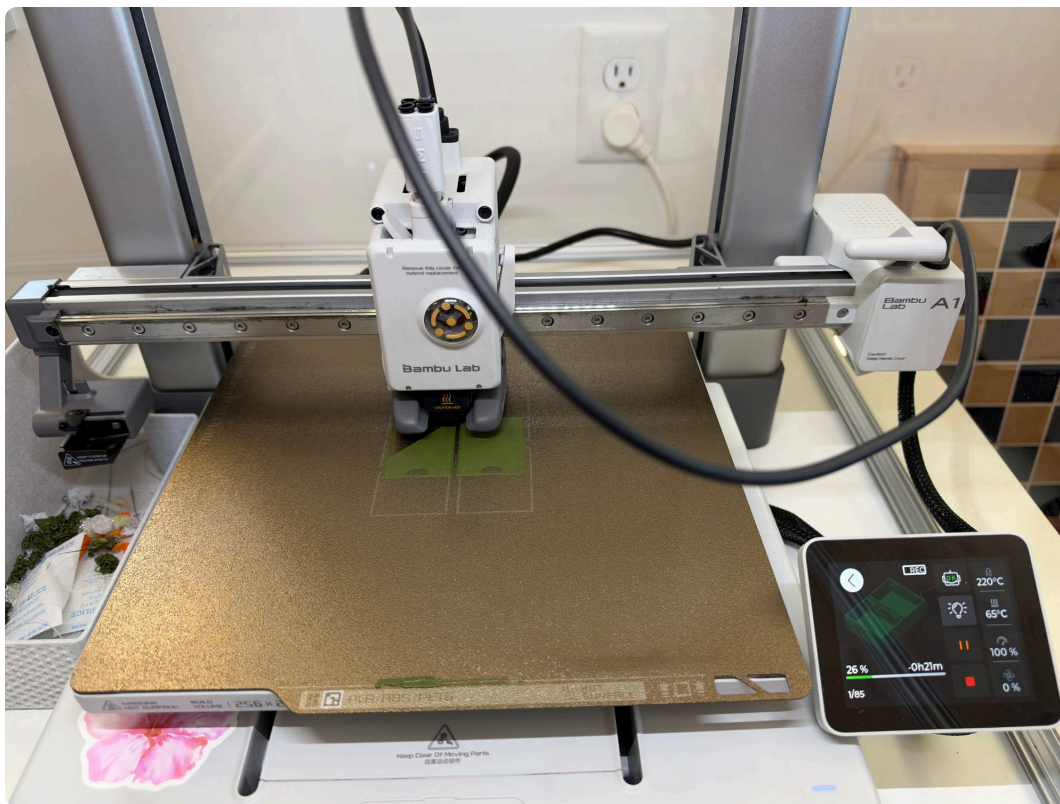
Raft layers	0	layers
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TO ADD: your reasons for the slicer settings, if they differ from the descriptions above.

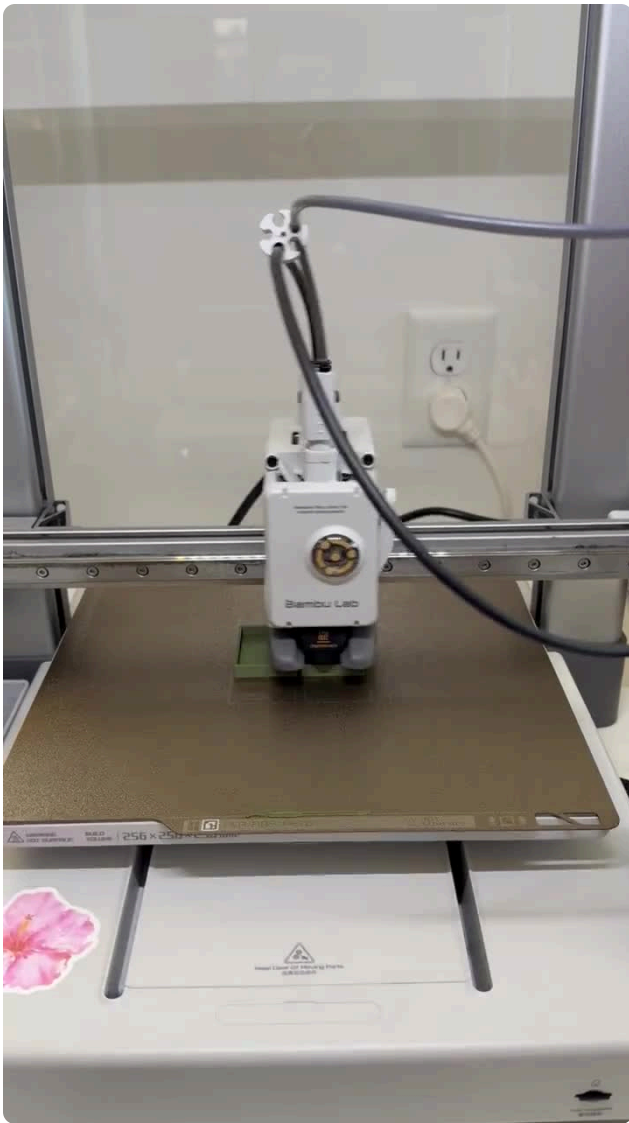
Slicing Result	
Color Scheme	▼ Filament
Filament	Model
1	4.62 m 13.55 g
Filament change times: 0	
Cost: 0.33	
Time Estimation	
Prepare and timelapse time: 6m15s + 2m16s	
Model printing time: 20m20s	
Total time: 28m52s	
Options	Display
Travel	
Retract	
Unretract	
Wipe	
Seams	☒

Slicing result: 4.62 m of filament, 13.55 g for both cases. Model printing time is 20 minutes 20 seconds, and the total time with preparation is 28 minutes 52 seconds.

Print Artifact

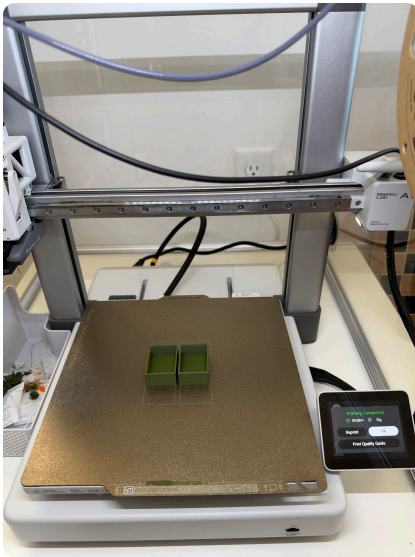


The print at 26 percent, with 21 minutes left. The nozzle is at 220°C and the bed is at 65°C.



Still frame from the print video. The full video is on the portfolio page.

Video of the two cases printing on the A1.



The finished print on the plate. The screen shows Printing Completed, 28 minutes and 13 g.

Fit Test

I printed the case and tested it with the speaker before class. The speaker fit on the first print, so I didn't need to modify the design after printing. The only changes I made were the ones I made before printing, when I settled on 0.2 mm of clearance.

TO ADD: photos of the speaker snapped into the printed case.

Show and Tell

The speaker snaps into the case and stays in place. The in-class test on the artifact is still to come. TO ADD: result of the in-class snap fit test.

Lessons Learned

Handbook tolerances are for machining: The tables in Machinery's Handbook assume machining, and the tolerances for an LC fit are far tighter than what an FDM printer can hold. I used the table as the design intent and picked the actual allowance from the printer's real accuracy.

Keep the allowance adjustable: Making the clearance a parameter meant I could change it in one place. I went from 0.2 mm to 0.4 mm and back to 0.2 mm without rebuilding anything.

Thin walls flex on their own: I planned relief cuts around the lips, but 1.6 mm walls already flex enough for the speaker to snap in, so I skipped them and the design stayed simpler.

TO ADD: mistakes you made along the way and how you fixed them.

TO ADD: actual time it took from start to finish.

Resources

Machinery's Handbook, 31st Ed., p. 640 (tolerance and allowance definitions) and p. 650 (Table 6, Standard Tolerances, and Table 7, Relation of Machining Processes to Tolerance Grades, ANSI/ASME B4.1-1967 (R2020))

Bambu Lab A1 3D printer, Bambu Studio slicer, SolidWorks, Overture Matte PLA, and a digital caliper (0.01 mm resolution)

Download

Speaker case (SolidWorks part): [L06-SpeakerBox.SLDPRT](#)

TO ADD: STL file of the case, if you want it included.